

组合电路

No.

Date. / /

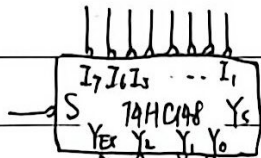
编码器

① 普通编码器

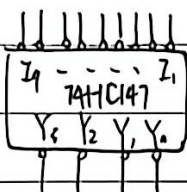
② 优先编码器 (二进制)

$$\begin{cases} Y_2 = I_4 + I_5 + I_6 + I_7 \\ Y_1 = I_2 + I_3 + I_6 + I_7 \\ Y_0 = I_1 + I_3 + I_5 + I_7 \end{cases}$$

$$\begin{cases} Y_2' = (I_4 + I_5 + I_6 + I_7)S' \\ Y_1' = (I_2 I_4' I_5' + I_3 I_4' I_5' + I_6 + I_7)S' \\ Y_0' = (I_1 I_2' I_4' I_6' + I_3 I_4' I_6' + I_5 I_6' + I_7)S' \\ Y_3' = (I_0' I_1' I_2' I_3' I_4' I_5' I_6' I_7' S')' \\ Y_{ex}' = (I_0 + I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7)S' \end{cases}$$



$S'=0$ 工作
 $Y_3'=0$ 无输入
 $Y_{ex}'=0$ 有



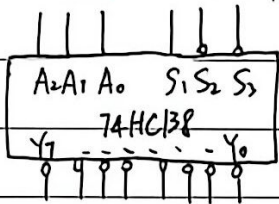
③ (二-十进制)

$$\begin{cases} Y_3' = (I_8 + I_9)' \\ Y_2' = (I_7 I_8' I_9' + I_6 I_8' I_9' + I_5 I_8' I_9' + I_4 I_8' I_9')' \\ Y_1' = (I_7 I_8' I_9' + I_6 I_8' I_9' + I_5 I_4' I_8' I_9' + I_2 I_4' I_5' I_8' I_9')' \\ Y_0' = (I_9 + I_7 I_8' I_9' + I_5 I_6' I_8' I_9' + I_3 I_4' I_6' I_8' I_9' + I_1 I_2' I_4' I_6' I_8' I_9')' \end{cases}$$

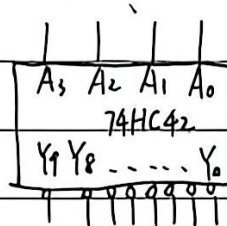
译码器

① 二进制译码器

② 二-十进制译码器



$$\begin{cases} Y_0' = (A_2' A_1' A_0') = m_0 \\ \vdots \\ Y_7' = (A_2 A_1 A_0) = m_7 \end{cases}$$



$$\begin{cases} Y_0' = (A_3' A_2' A_1' A_0') \\ \vdots \\ Y_9' = (A_3 A_2 A_1 A_0) \end{cases}$$

$A_2 A_1 A_0 = 000$

$\Rightarrow Y_0 = 0$

$S_1=0$ 不工作
 $S_1=1 \quad S_2+S_3=0$ 正常工作
 $S_1=X \quad S_2+S_3=1$ 不工作

△ 关于取反

① 编码器的输入/输出/控制

端均取反✓

② 译码器 输入 X 输出✓

$S_1 X \quad S_2 S_3$ ✓ 控制端✓

③ 数据选择器 输入/输出 X

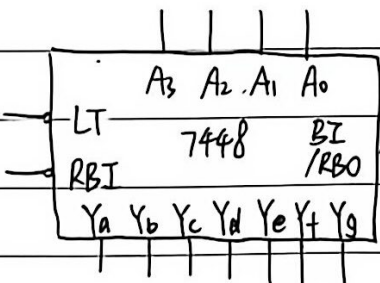
S 取反

④ 加法器

⑤ 比较器

全不取反

② BCD-七段显示译码器



灯测试输入 $LT'=0$ (灯测试全亮)

灭零输入 $RBI'=0$

灭灯输入 $BI'=0$ / 灭零输出 $RBO'=0$



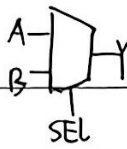
CS 扫描全能王

3亿人都在用的扫描App

No.

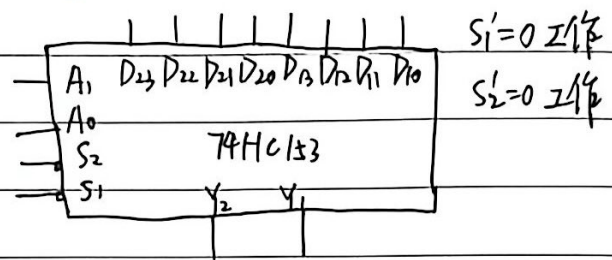
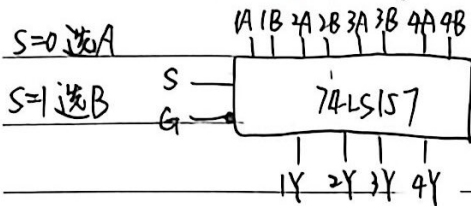
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$$Y = \text{SEL} \cdot A + \text{SEL}' \cdot B$$



数据选择器 ① 图2选1

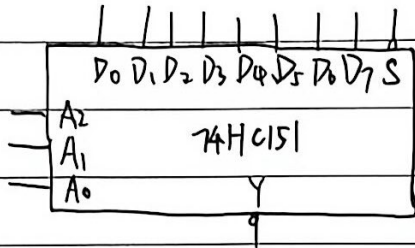
② 双4选1



③ 8选1

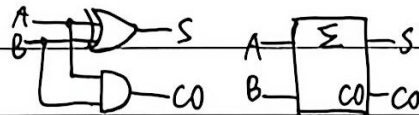
$S'_1 = 0$ 工作

$$Y_1 = [D_0(A_1A_0)' + D_1(A_1A_0)' + D_2(A_1A_0)' + D_3(A_1A_0)] \cdot S_1$$



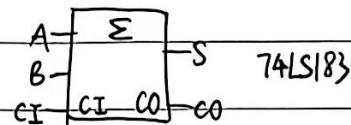
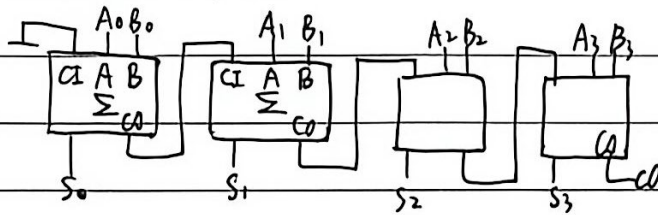
加法器 ① 半加器

② 全加器

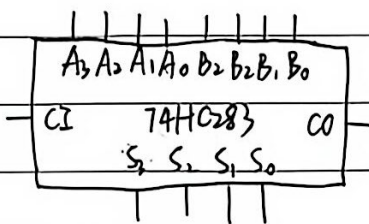


$$\begin{cases} S = (A'B'CI' + AB'CI + A'B'CI + ABCI)' \\ CO = (A'B' + B'CI' + A'CI)' \end{cases}$$

③ (行波) 串行进位加法器



④ (快速) 超前进位加法器 每位进位不用等前一位, 直接从 A_iB_i 和 CI 算出



$$(CO)_i = A_iB_i + (A_i + B_i)(CI)_i = G_i + P_i(CI)_i$$

$$= G_i + P_iG_{i-1} + P_iP_{i-1} [G_{i-2} + P_{i-2}(CI)_{i-2}]$$

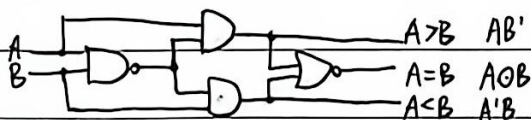
$$S_i = A_iB_i(CI)_i' + A_i'B_i(CI)_i + A_iB_i(CI)_i' + A_iB_i(CI)_i$$

$$= A_i \oplus B_i \oplus (CI)_i$$

数值比较器

1位

多位



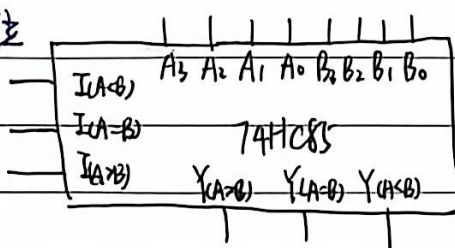
$$\begin{aligned} Y_{(A>B)} &= A_3B_3' + (A_3 \odot B_3)A_2B_2' + (A_3 \odot B_3)(A_2 \odot B_2)A_1B_1' \\ &\quad + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)A_0B_0' \\ &\quad + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)I_{(A>B)} \end{aligned}$$

$$Y_{(A=B)} = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)I_{(A=B)}$$

$$Y_{(A>B)} = (Y_{(A<B)} + Y_{(A=B)})'$$

从高位到低位

优先级



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